FrameContact: sequence-predicted geometry and seed-matched task routing

Antibody sequence

EVQLVESGGGLVQPGGSLR...

PLM cache ESM-2 650M 1280d → 128d

Predicted geometry N / CA / C cache

Antigen sequence

MKTIILASYIFCLVFADY...

PLM cache ESM-2 650M 1280d → 128d

Predicted geometry N / CA / C cache

Interleaved Geometry-Aware Partner Exchange Block

× 2

Symmetric low-rank pair decoder

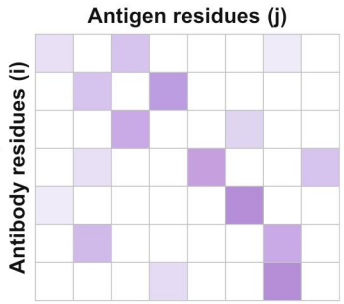
$\varphi(x,y) = q(x)^T k(y)/\sqrt{d_p} + u(x) + v(y) + b$

$z_{ij} = \tfrac{1}{2}[\varphi(h_i^{Ab}, h_j^{Ag}) + \varphi(h_j^{Ag}, h_i^{Ab})]$

$p_{ij} = \sigma(z_{ij}), \quad d_p = 128$

Symmetric Low-Rank Pair Decoder

Pair probabilities p^r



SEQUENCE-ONLY INPUT BOUNDARY

- PLM residue embeddings
- chain-internal predicted N/CA/C
- no cross-partner coordinates or true distances
- no contact labels as model input

Interleaved Geometry-Aware Partner-Exchange Block (one of L = 2)

Parallel Ab / Ag intra-chain encoders

Dense content self-attention
‖ Local geometry-biased self-attention
same physical chain · RoPE · SiLU FFN
$H = H_{dense} + \tanh(g_{geom})(H_{local} - H_{in})$;
$g_{geom} = 0$ init

Simultaneous partner exchange

Ab ← Ag
Ag ← Ab
multi-head · content-only
both directions read the same pre-cross states

Per block: [parallel Ab/Ag intra-chain encoding] → [simultaneous Ab←Ag and Ag←Ab]

Geometry enters only the local same-chain attention bias; pair/contact logits are not used by the frozen content-only cross-attention.

Seed-matched fixed router $r \in \{17, 29, 43\}$: contact + paratope ← J_r | epitope ← E_r

CONTACT · J_r
pairwise output
$p^c_{ij} = \sigma(z^J_{ij})$

PARATOPE · J_r
direct Ab node logit
+ masked LME_j(z^J_{ij})
→ sigmoid

EPITOPE · E_r
for each Ag residue j:
top-2 valid $\sigma(z^E_{ij})$
mean → clip → logit

Continuous scores are routed; no mixture-weight tuning and no thresholded-label routing.